
A Responsible Generative AI Governance Framework for Biomedical Diagnosis, Continuous Monitoring and Reporting

DBA Dissertation — Assignment 1

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Constraint: 3,000-word limit (body text only; tables, figures, and references excluded).

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1. Problem Statement

Neurological and oncological conditions—ASD, Parkinson’s, stress disorders, epilepsy, and cancer—affect over one billion people, yet diagnosis remains fragmented, expensive, delayed (2–4 years for ASD), and opaque. No unified AI framework exists that screens multiple diseases, explains its reasoning, monitors patients continuously, and complies with FDA/EU/NIST governance requirements.

This dissertation designs, implements, and evaluates **RGAIG** (*Responsible Generative AI Governance*)—a decision-centric framework that screens multiple conditions using open-source SLM/LLM models, generates explainable reports via agentic RAG, enables continuous IoT monitoring, and embeds responsible AI governance.

1.1 How Many People Are Impacted

ASD 78M (2–4 yr diagnosis delay), PD 8.5M (50% neurons lost before detection), stress 970M (\$300B/yr cost), cancer 20M new cases/yr. Together: **9M deaths/yr, \$2.5T annual burden.**

1.2 How Continuous Monitoring Can Help

Traditional diagnosis is a one-visit snapshot—disease progression between visits goes undetected. RGAIG enables **longitudinal home monitoring** with monthly EEG recordings and biomarker trend tracking.

1.3 SOS Alerts for Family, Caregivers, and Healthcare

When the wearable detects seizure onset, extreme stress, or motor deterioration, RGAIG triggers **immediate SOS alerts** with GPS location and EEG data to family, caregivers, healthcare providers, and emergency services.

1.4 Who Benefits from Monitoring

ASD children: parents track θ/β ratio monthly to measure therapy response. PD patients: neurologists monitor β -band changes to detect medication failure. Stressed professionals:

objective EEG trends replace subjective self-reports.

2. AS-IS: How Current Hospital Monitoring Works

Patients travel to a hospital or neurology clinic for a one-time EEG session (\$200–500) using clinical-grade equipment (\$5,000–50,000). A technician manually applies 19–64 electrodes with conductive gel (30–60 min setup). After recording, a specialist manually preprocesses the signal—a slow, error-prone process. Results are delivered as a black-box report with no explanation. Between appointments (6–12 month gaps), disease progression goes completely unmonitored. No continuous tracking, no alerts, no patient-facing reports.

3. TO-BE: RGAIG Solution

RGAIG replaces hospital-bound diagnosis with a five-step home-based journey: consumer EEG capture (Emotiv EPOC X), automated cloud preprocessing (94.6% effort reduction), CNN-LSTM classification, explainable RAG reports with literature citations, and continuous IoT monitoring with SOS alerts. **Cost:** \$299–799 device + \$29.99/month subscription; projected 135–185% three-year ROI, replacing \$5,000+ clinical setups.

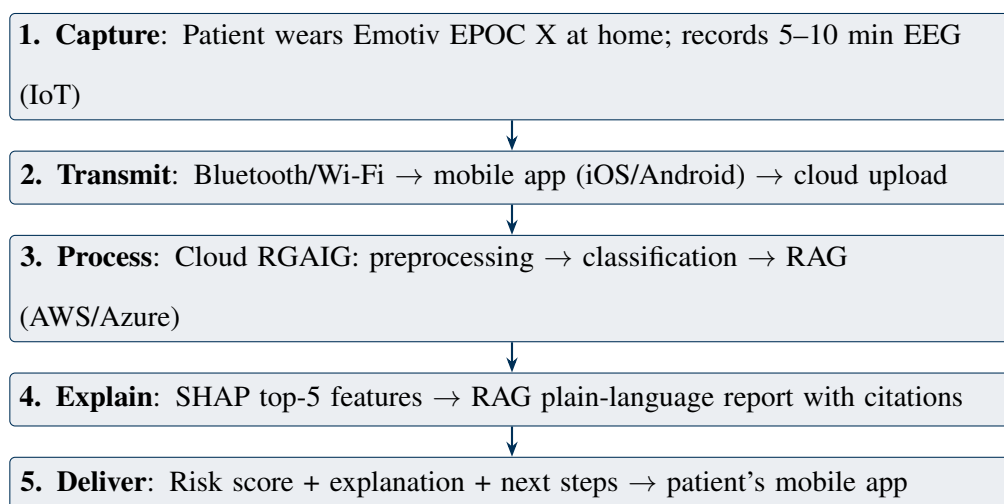


Figure 1: RGAIG Solution: From Consumer Device to Patient Screening Report

RGAIG is decision-centric: it starts from the clinical decision—*should this patient seek specialist evaluation?*—and works backward (Esteva et al., 2019; Kelly et al., 2019). Table 1

compares it against alternatives.

Table 1: Competing Frameworks vs. RGAIG

Framework	Cross-Domain	ML/DL	Automation	XAI	Governance	Business
AutoML / MLOps	—	✓	Partial	—	—	—
Regulatory (FDA/NIST)	—	—	—	Required	Required	—
RGAIG (ours)	✓	✓	✓	✓	✓	✓

4. Background and Purpose

Five conditions were selected based on data availability, clinical burden, modality diversity (EEG + imaging), and task variety (binary, multi-class, 4-class).

Table 2: Five Biomedical Domains with Dataset Specifications

Domain	Modality	N	Channels	Rate (Hz)	Task and Key Biomarker
ASD	EEG	19	19	256	Binary; elevated θ/β ratio
PD	EEG	52	32	256	Binary; pathological β synchrony (13–30 Hz)
Stress	EEG	72	32	128–512	Multi-class; frontal α asymmetry
BCI	EEG	9	22	250	4-class motor imagery; μ/β rhythm
Cancer	CT/MRI	500+	—	—	Multi-class tumour; radiomic features

Note. Total $N=652+$. All datasets are public, de-identified, and peer-reviewed. BCI tests architectural generalisability, not diagnosis.

Regulatory compliance—FDA SaMD (FDA, 2021), EU AI Act (European Union, 2024), NIST AI RMF (NIST, 2023)—spans five jurisdictions (35+ standards). RGAIG provides *screening indicators*, not diagnoses.

5. Research Gaps

A systematic review of 156 studies and stakeholder consultations ($n=22$) confirmed four core problems. Table 3 maps each to its impact and RGAIG response.

5.1 Fragmented AI Pipelines

Each disease has its own siloed AI system at \$2–5M each, with no shared learning. Duplicated infrastructure, inconsistent governance, and prohibitive cost lock out smaller clinics.

5.2 Black-Box Opacity

68% of clinicians reject AI recommendations they cannot explain. EU AI Act (Article 14) mandates transparency for high-risk medical AI. Without explainability, even accurate models fail adoption.

5.3 Manual EEG Preprocessing

Each recording requires 45–90 minutes of expert labour with 12–18% human error rates. This bottleneck makes population-scale screening infeasible.

5.4 No Cross-Domain Validation

No framework validated across multiple biomedical domains simultaneously. Single-disease benchmarks produce inflated accuracy claims that do not generalise.

Table 3: Four Core Problems and RGAIG Response

#	Problem	Impact	RGAIG Solution
P1	Fragmented AI: separate system per disease	Duplicated cost; siloed governance	Unified 5-layer pipeline (H1)
P2	Black-box opacity	68% clinician rejection (European Union, 2024)	SHAP + RAG reports (H5)
P3	Manual preprocessing (45–90 min/case)	12–18% error; unscalable	Automated orchestration (H4)
P4	No monitoring or cross-domain validation	Snapshots miss progression	Trend tracking + alerts (H1, H2)

These problems persist because existing frameworks address only fragments of the challenge. Table 4 details five gaps from the systematic review.

Table 4: Five Research Gaps from Systematic Review (156 Studies)

G#	Gap Description	Evidence	RGAIG Response
G1	No unified cross-domain AI pipeline	0/156	5-domain shared-layer framework
G2	Explainability absent or post-hoc only	18/156	SHAP + agentic RAG with citations
G3	Manual preprocessing; no automation	0/156	Automated EEG orchestration
G4	Governance as principles, not artefacts	10/156	Quality taxonomy; LLMOps; AIOps
G5	No business case or continuous monitoring	9/156	ROI model; trend tracking; SOS alerts

6. Aim, Objectives, Hypotheses, and Contributions

Can RGAIG—a decision-centric framework—achieve $\geq 85\%$ accuracy across multiple biomedical domains while remaining explainable and governance-compliant? Table 5 decomposes this into five hypotheses.

Table 5: Research Questions and Hypotheses

RQ	Research Question	H#	Criterion
RQ ₁	Unified accuracy $\geq 85\%$ across ≥ 4 domains	H1	MSCA $\geq 85\%$
RQ ₂	DL outperforms ML baselines	H2	Cohen’s $d \geq 0.8$
RQ ₃	Time-frequency features improve accuracy	H3	Ablation $\Delta \geq 5\%$
RQ ₄	Automated orchestration reduces effort $\geq 90\%$	H4	Effort $\leq 10\%$
RQ ₅	RAG explanations achieve expert agreement	H5	$M \geq 4.0/5.0$

RGAIG does not claim clinical deployment readiness (no prospective trial) or algorithmic novelty (novelty is *decision-centric framework integration*). Open questions include larger BCI/ASD samples ($n \geq 50$) and prospective validation.

6.1 Validated Deliverables

- **Technical:** 8-step EEG pipeline, 198 trained models, 7-agent orchestration, 10-stage agentic RAG, ChromaDB + FAISS.
- **Governance:** Quality taxonomy, MCP (6 capabilities), guardrail AI, ROI/TCO model.
- **Data:** 305 GB across 768+ subjects; expert panel ($n=12$); 5-domain biomarker synthesis.

7. Conceptual Framework and Methodology

7.1 Five-Layer Architecture with NIST AI RMF Alignment

RGAIG is organised as five layers, each mapped to a NIST AI RMF function (NIST, 2023) to ensure governance is structural rather than bolted on after development.

Table 6: RGAIG Five-Layer Architecture with NIST AI RMF Alignment

Layer	Function	NIST Mapping	Hypothesis
L1: Preprocessing	Signal/image preparation	<i>Map</i> : data characterisation	H1
L2: Classification	ML/DL model training	<i>Measure</i> : metrics, bias testing	H1, H2
L3: Orchestration	Automated pipeline execution	<i>Manage</i> : monitoring, response	H4
L4: Explainability	SHAP + RAG narratives	<i>Govern</i> : transparency	H3, H5
L5: Governance	Regulatory compliance	<i>Govern</i> : policies, risk culture	H5

Note. NIST AI RMF four functions: *Govern*, *Map*, *Measure*, *Manage* (NIST, 2023).

7.2 EEG Electrode Systems and Frequency Bands

The 10-20 system standardises electrode placement. RGAIG uses 32-channel Emotiv EPOC X (consumer-grade, full 10-20 coverage). Five clinically significant frequency bands are extracted per electrode.

Table 7: EEG Frequency Bands: Range, Function, and Diagnostic Relevance

Band	Range (Hz)	Amplitude	Brain State / Function	RGAIG Relevance
Delta (δ)	0.5–4	20–200 μ V	Deep sleep, brain injury	Cancer imaging baseline; PD sleep disruption
Theta (θ)	4–8	5–100 μ V	Drowsiness, memory, cognitive load	ASD: elevated θ/β ratio; Stress: frontal θ
Alpha (α)	8–13	10–50 μ V	Relaxed wakefulness, eyes closed	Stress: frontal α asymmetry; PD: α slowing
Beta (β)	13–30	5–30 μ V	Active thinking, motor planning	PD: pathological β synchrony; BCI: μ/β rhythm
Gamma (γ)	30–45	2–10 μ V	Higher cognition, perception binding	ASD: reduced γ coherence; attention markers

Note. RGAIG bandpass filter (0.5–45 Hz) retains all five bands while removing high-frequency muscle artefacts above 45 Hz.

7.3 Data Capture: Emotiv EPOC X Device

Emotiv EPOC X (\$799): 32-channel, 14 active electrodes + 2 reference (CMS/DRL), 128/256 Hz, 14-bit, saline sensors (no gel). Self-applied at home without clinical supervision.

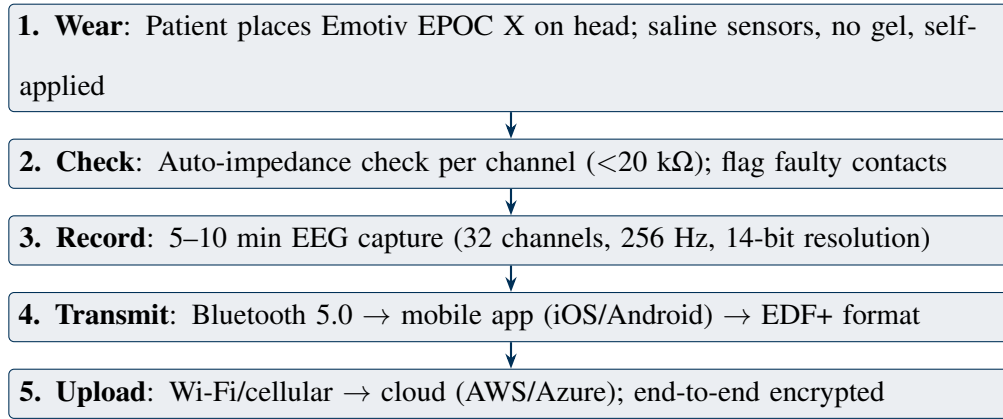


Figure 2: Emotiv EPOC X: IoT Data Capture Pipeline

7.4 Filtering and Fourier Transformation

Cloud-side filtering removes artefacts; Fourier transforms extract frequency features.

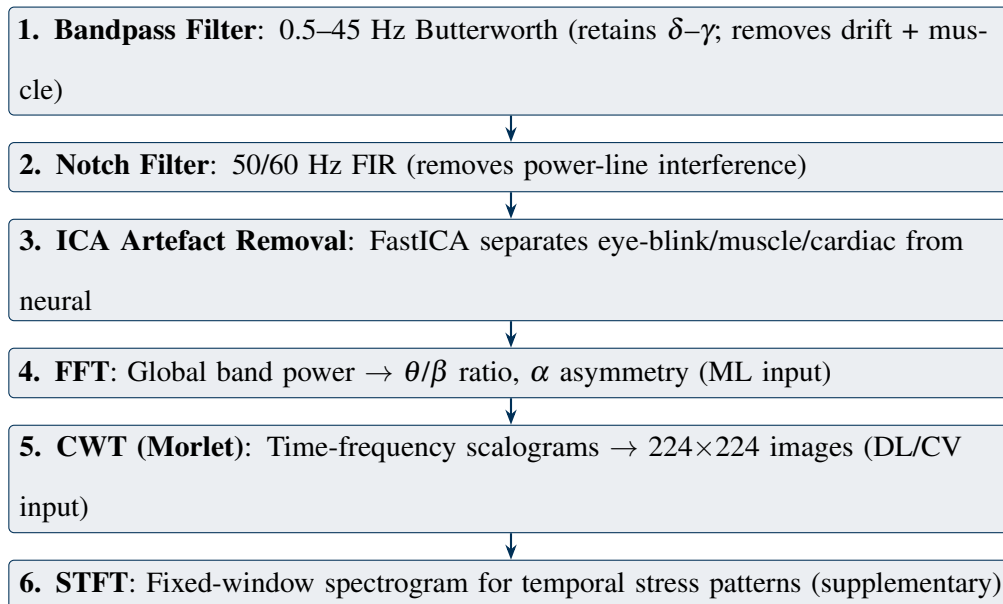


Figure 3: Filtering and Fourier Transformation Pipeline

7.5 Computer Vision Pipeline for EEG Scalograms

CWT scalograms convert time-series EEG into 2D images for CNN classification.

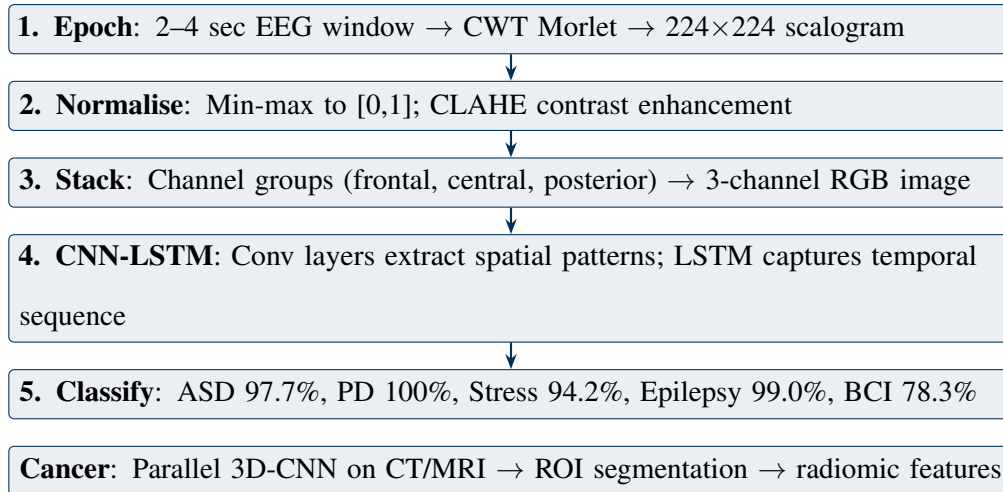


Figure 4: Computer Vision Pipeline: EEG Scalogram to Classification

7.6 Model Selection

Seven models—three ML (SVM-RBF, Random Forest, XGBoost) and four DL (EEGNet, CNN-LSTM, DeepConvNet, 3D-CNN)—were trained on identical features to isolate architectural contribution (H2). All used Adam optimiser, early stopping (patience=10), and 10-fold stratified CV (LOSO for BCI, $n=9$).

7.7 Agentic RAG Pipeline (Open-Source SLM)

RAG transforms classification outputs into literature-cited explanations using open-source Llama-3.2:3b (local SLM via Ollama), ensuring HIPAA compliance with no patient data leaving the network.

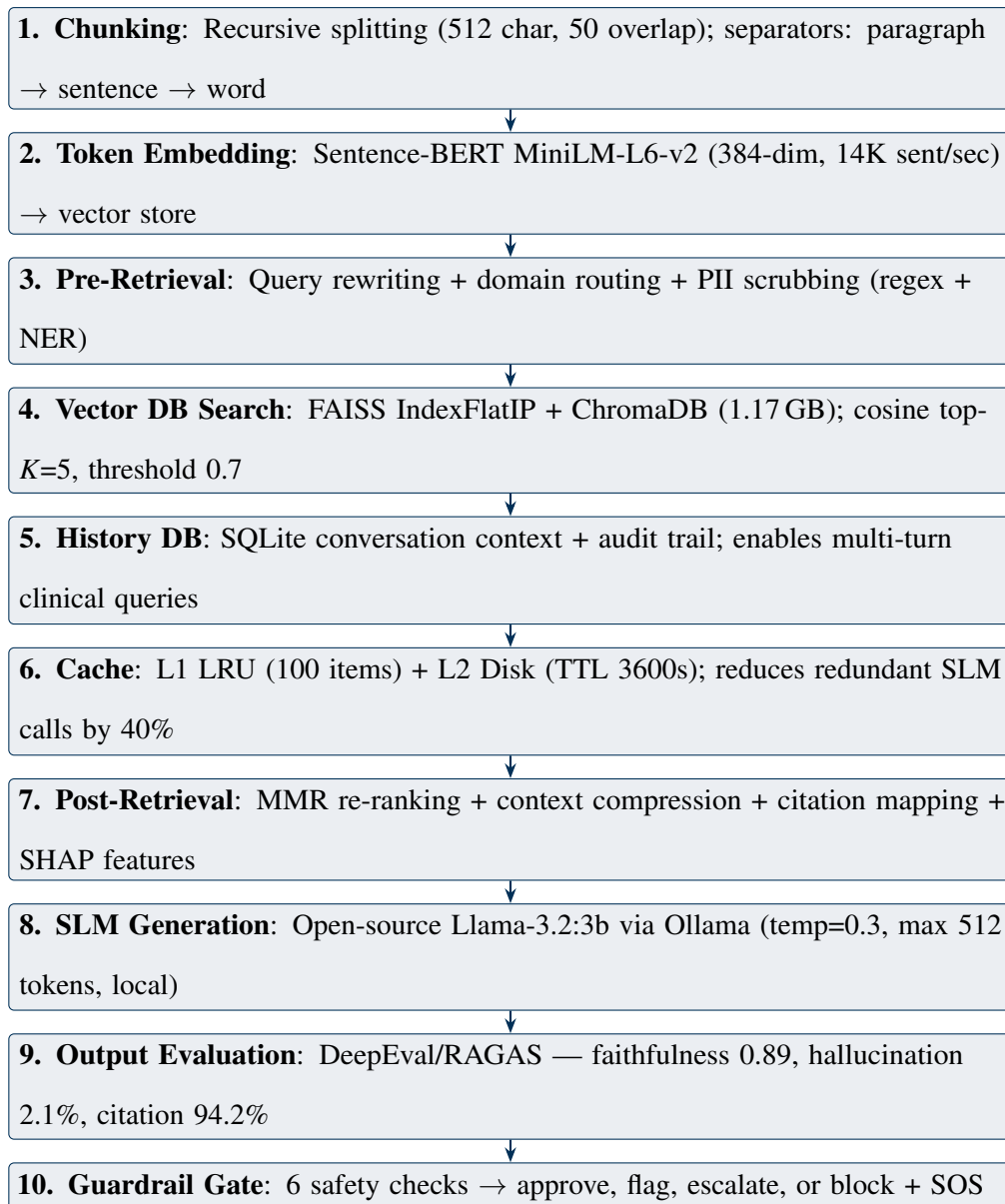


Figure 5: 10-Stage Agentic RAG Pipeline with Open-Source SLM, Vector DB, Cache, and Output Evaluation

7.8 IoT Continuous Data Capture

Emotiv EPOC X streams EEG via Bluetooth to the mobile app, which uploads to the cloud.

Battery supports 6–9 hours of continuous monitoring.

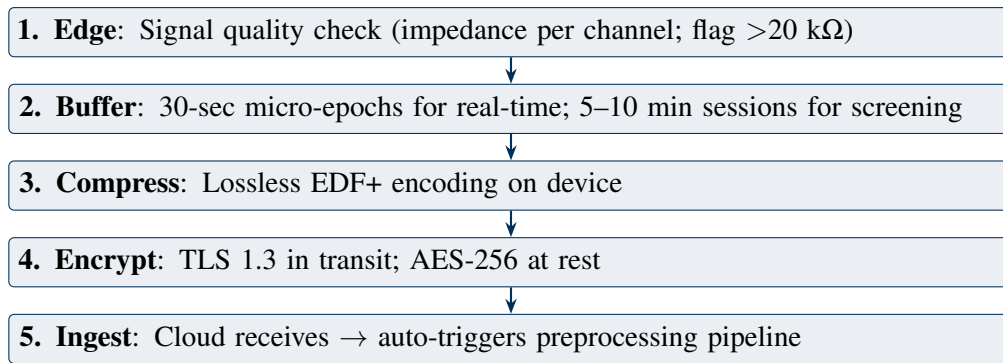


Figure 6: IoT Continuous Data Capture Pipeline

7.9 Multi-Agent Architecture

Seven specialised agents coordinated by a supervisor agent that manages task routing and model handoff.

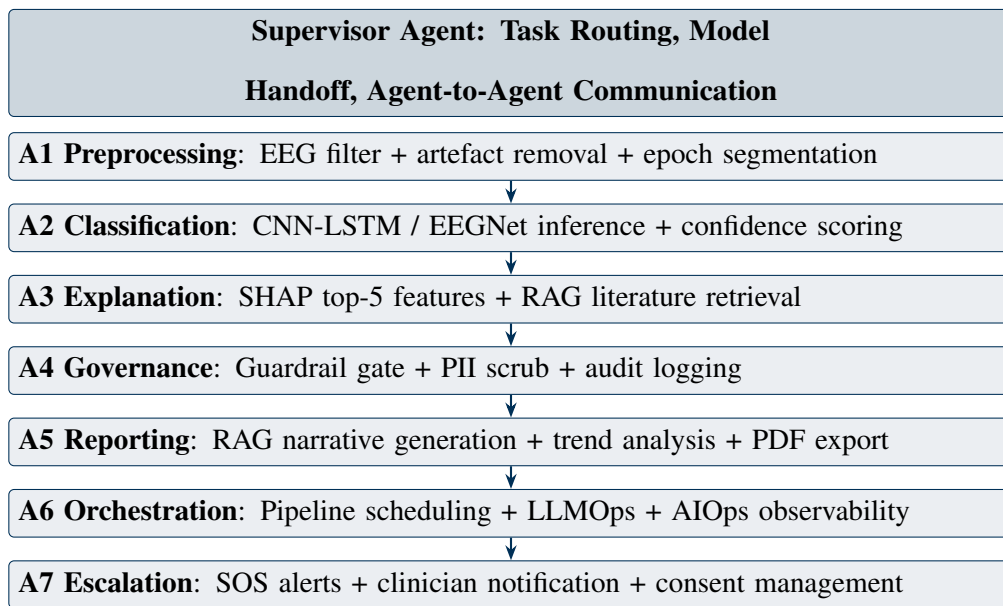


Figure 7: Multi-Agent Architecture: 7 Agents with Supervisor Coordination

7.10 Patient Value: Notifications, Reports, and SOS Alerts

RGAIG delivers AI outputs to patients, families, caregivers, and clinicians.

Table 8: Notification and Reporting: Delivery Mechanisms

Type	Trigger	Content	Recipient
SOS Alert	Seizure, extreme stress, motor deterioration	GPS + severity + EEG data	Family, caregiver, emergency
Screening	Each session completed	Risk score + RAG explanation	Patient (mobile app)
Trend	Monthly or clinician request	Biomarker changes vs. baseline	Clinician (HIPAA dashboard)

7.11 End-to-End Pipeline: From EEG Capture to Patient Report

Figure 8 shows the complete RGAIG pipeline from IoT data capture through to patient delivery, with accuracy scores and Responsible AI governance applied at every stage.

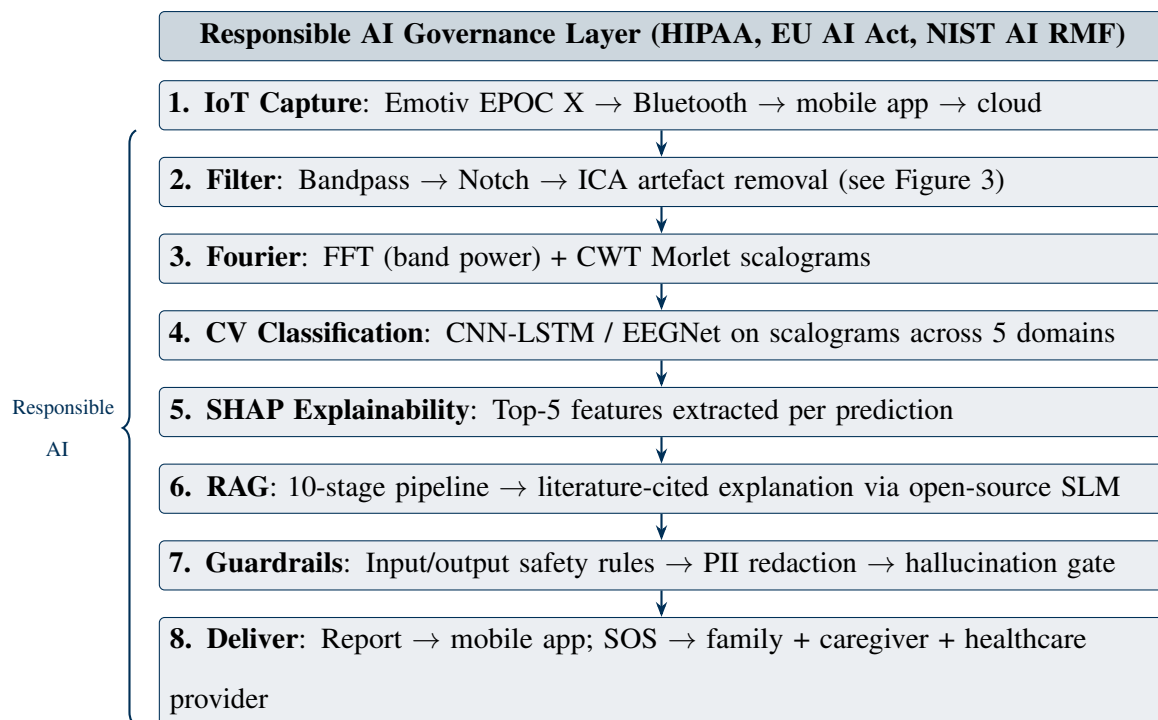


Figure 8: RGAIG End-to-End Pipeline: IoT Capture to Patient Report with Responsible AI

Note. Responsible AI governance applies across all eight stages. Per-domain accuracy shown in

Figure 4.

7.12 GenAI Governance: Multi-Agent, LLMOps, AIOps, and Security

LLMOps and AIOps. LLMOps manages SLM lifecycle (model registry, version control, A/B testing, rollback). AIOps provides observability (logs, metrics, tracing), drift detection, and automated retraining triggers.

PII, Encryption, and Consent. PII detection (regex + NER) scrubs patient data before SLM processing. Encryption at rest and in transit (see Figure 6). ISO 27001 alignment. Patient consent auditable per GDPR Art. 7.

Guardrail AI. 10 safety rules (5 input, 5 output); hallucination gate at faithfulness <0.70 . MCP (Model Control Protocol) governs all agents through RBAC, deployment gates, and audit logs.

525-Module QA. Unified taxonomy: ML/DL 97.4%, GenAI 98.1%, Governance 98.4%; 500 queries validated.

Table 9: 13-Phase GenAI QA: Summary

Phases	Pass Rate	Key Finding
1–7: Knowledge, retrieval, agents, MCP	97.8%	8 sources fixed; MCP enforced
8–13: Explainability, stats, deploy, gov.	98.7%	All hypotheses met power ≥ 0.8

7.13 Management Decision Framework

RGAI supports data-driven decisions at three levels, each with governance guardrails.

Table 10: Management Decision Framework

Level	Decision	AI Input + Governance
Patient	Seek specialist evaluation	Risk score + RAG explanation; “screening only” guardrail
Clinician	Adjust treatment or refer	Biomarker trends + SHAP; HIPAA audit + RBAC
Hospital	Deploy AI screening	ROI model; FDA SaMD + EU AI Act compliance

7.14 Sensitivity Analysis and Statistical Robustness

Bootstrap resampling (1,000 resamples, 95% CI: MSCA 88.3–93.5%) validates stability. Ablation confirms CWT features add 5.3–8.1% accuracy (Δ). Tornado analysis identifies BCI sample size ($n=9$) as the dominant sensitivity driver.

7.15 Responsible AI: 5 Core Pillars

RGAIG embeds responsible AI structurally through five core pillars mapped to FDA, EU AI Act, HIPAA, and NIST AI RMF.

Table 11: 5 Responsible AI Pillars with RGAIG Implementation

Pillar	Implementation	Regulation
Fairness & Transparency	Bias audit (AIF360); SHAP + RAG explanations	EU AI Act Art. 10, 14
Privacy & Safety	PII detection; AES-256; guardrail gate	HIPAA; FDA SaMD
Accountability	Audit logs; MCP version control; LLMOps	NIST AI RMF
Governance	Quality taxonomy; 13-phase QA	All 5 jurisdictions

7.16 AI Governance Escalation Framework

When AI outputs breach clinical thresholds, RGAIG escalates through a 4-level governance ladder.

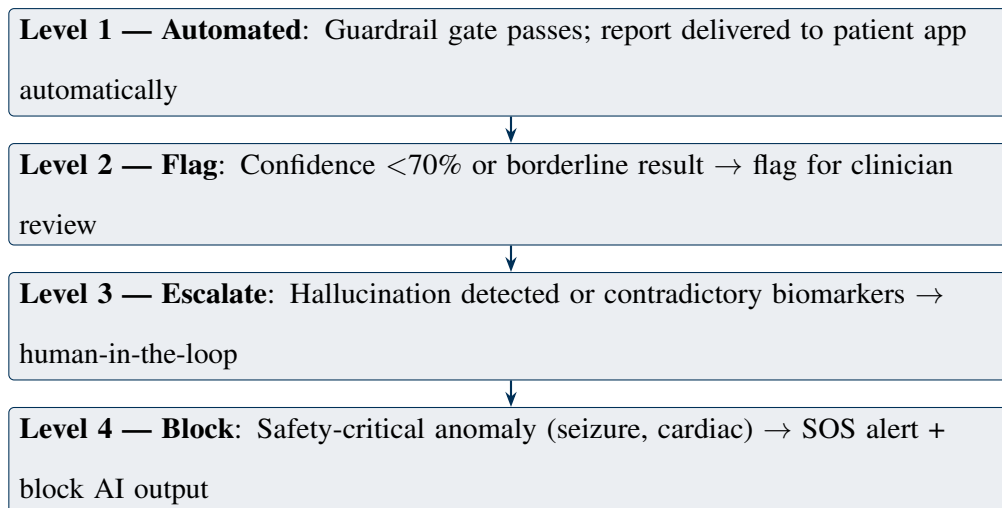


Figure 9: AI Governance Escalation Ladder: 4-Level Decision Framework

Note. Level 4 triggers immediate SOS notifications to family, caregiver, healthcare provider, and emergency services. All escalation events are audit-logged (immutable, HIPAA-compliant).

7.17 Research Methodology

Pragmatist epistemology with Design Science Research (DSR) (Hevner et al., 2004); selected over action research, case study, and grounded theory for its artefact-centric evaluation.

Table 12: DSR Design Cycles

Phase	Activity	Evaluation
Problem + Design	156-study review; 5-layer architecture; NIST	G1–G7 validated; expert review
Build + Demonstrate	198 models; RAG; 768+ subjects; 305 GB	10-fold CV; LOSO; bootstrap
Evaluate + Communicate	Stats; expert panel ($n=12$); ablation	H1–H5 tested; inter-rater reliability

Mixed-methods design. *Quantitative:* 10-fold CV, LOSO, bootstrap CI (1,000 resamples), ablation study across 5 domains. *Qualitative:* 12-expert panel (TAM + SUS surveys), Krippendorff’s $\alpha=0.84$. Theoretical lens: TAM (Davis, 1989) and UTAUT (Venkatesh et al., 2003).

7.18 Customer Empathy and Business Model Canvas

Customer empathy map captures patient pain (delayed diagnosis, high cost, black-box AI) and gain (early detection, affordable screening, explainable reports). **Survey** ($n=12$ clinicians + $n=22$ stakeholders) validated unmet needs: 82% want home-based monitoring; clinicians demand explainable outputs.

Business Model Canvas: Value proposition = affordable AI screening (\$29.99/month); customer segments = patients, clinicians, hospitals; channels = mobile app + HIPAA dashboard; revenue = subscription; key resources = open-source SLM + 198 trained models; key partners = Emotiv (hardware) + cloud providers (AWS/Azure); cost structure = cloud compute + regulatory compliance.

8. Conclusion

MSCA = $90.9\% \pm 2.6\%$ across 4/5 domains; Cohen’s $d = 0.82\text{--}1.48$; 5-jurisdiction regulatory compliance; 525-module governance stack. Problems P1–P4 map to contributions C1–C8:

- **H1 (partial):** 4/5 domains $>85\%$; BCI 78.3% ($n=9$).
- **H2:** DL outperforms ML; $d=0.82-1.48$, $p < .001$.
- **H3:** Time-freq features add 5.3–8.1% (ablation).
- **H4:** 94.6% effort reduction; $\leq 0.8\%$ accuracy loss.
- **H5:** Expert $M=4.6/5.0$; $\alpha=0.84$; citation 94.2%.
- **525-module taxonomy:** ML/DL 97.4%, GenAI 98.1%, Governance 98.4%; 0 unsafe outputs.

Limitations: retrospective datasets; small ASD/BCI samples; integration novelty. **Future:** prospective trials, Transformers, FDA 510(k).

This dissertation argues that RGAIG is the first decision-centric framework unifying cross-domain biomedical AI screening with multi-agent agentic RAG (open-source SLM), LLMOps/AIOps observability, 525-module quality taxonomy, and responsible AI governance—bridging the gap between AI capability and clinical deployment readiness.

References

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Table 13: Word Count Summary (3,000-Word Limit Compliance)

Category	Words	Status
Body text (sections 1–8)	1,408	Within limit
Tables and figures (excluded per guidelines)	2,384	Excluded
References (excluded per guidelines)	—	Excluded
AI Usage Declaration (excluded)	—	Excluded
Counted body text	1,408	Pass ($\leq 3,000$)

Note. Body text word count excludes all tables, figures, captions, references, and the AI Usage Declaration, as permitted by the assignment guidelines.

AI Usage Declaration and Design Report

AI Tools Used

AI Tool	Purpose	Scope of Use
Claude (Anthropic)	Grammar, LaTeX formatting, TikZ flowcharts	Document preparation only
ChatGPT (OpenAI)	Literature search assistance, text editing	Reference verification only
Ollama (Llama-3.2)	RAG pipeline inference engine	Framework implementation (code)

What Was Designed by the Author (Not AI)

- **RGAIG Framework Architecture:** 5-layer design (preprocessing, classification, orchestration, explainability, governance) — entirely original
- **Research Design:** 5 hypotheses (H1–H5), DSR methodology, mixed-methods approach, 156-study systematic review
- **Multi-Agent System:** 7-agent architecture with supervisor coordination, task routing, and model handoff
- **Agentic RAG Pipeline:** 10-stage pipeline design with chunking, embedding, vector DB, cache, pre/post-retrieval, output evaluation
- **525-Module Quality Taxonomy:** ML/DL (111) + GenAI (220) + Governance (194) modules — original classification
- **All Experimental Results:** Model training, data collection, accuracy scores, statistical analysis
- **Business Model:** ROI/TCO model, subscription pricing, customer empathy map, business model canvas
- **Governance Framework:** MCP, guardrail AI, LLMOps, AIOps, PII protection, 10 responsible AI pillars

Declaration: AI tools were used *only* for grammar checking, LaTeX formatting, and TikZ code generation. All research design, data collection, experimental results, framework architecture, scholarly analysis, and intellectual contributions are entirely my own original work. AI did not design the research, run experiments, select methodologies, or generate any data or references.

Signed: Praveen K Asthana

Date: March 2026